



Get the poster
janiserdmanis.org

Verifiable Cleansing via Blinded Credential Grouping

E2E Verifiable · Everlasting Privacy · Update Deniability · Lightweight Smartcard

Janis Erdmanis^{1,2}

¹ Independent Researcher

² Seeking a Postdoctoral Position

2 Verifiable Cleansing Construction (Main Result)

The collector matches a shuffle of the M accepted ballots against a blinded shuffle of the K -entry registration roll, groups entries by blinded credential into N groups and proves that each selected entry is the most recently recorded ballot. D dummies pad the groups to the same size (3 here).

$$\text{PoK} \left\{ \psi : E'_i = \text{Com}(E_{\psi(i)}) \wedge R'_i = \text{Com}(g^{R_{\psi(i)}}) \wedge V'_i = \text{ReEnc}(V_{\psi(i)}) \right\}$$

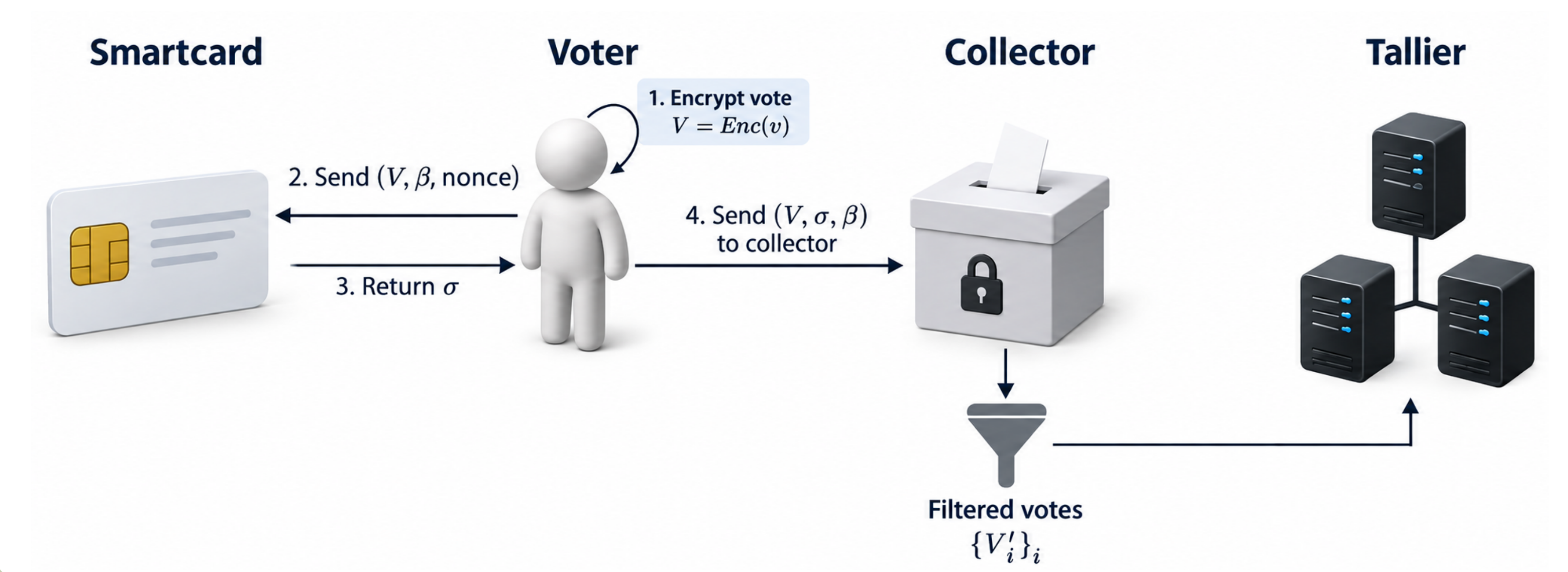
$$\text{PoK} \left\{ \phi : E''_k = \text{Com}(\text{pk}_{\phi(k)}) \wedge \{E'_i\}_{i \in L} \subseteq \{E''_k\}_k \right\}$$

	R	E	V	R'	E'	V'		E''	pk	
π_{E_1}	0	1	1	R'_{11}	E'_{11}	V'_{11}	$\pi_b \quad \pi_{gt}$	E''_1	pk_1	Alice
	0	1	1	R'_{12}	E'_{12}	V'_{12}		E''_2	pk_2	Bob
	$\vdots$	$\vdots$	$\vdots$	R'_{13}	E'_{13}	V'_{13}	$\pi_b \quad \pi_{gt}$	E''_3	pk_3	Eve
	0_D	1_D	1_D	R'_{21}	E'_{21}	V'_{21}	$\pi_b \quad \pi_{gt}$	E''_4	pk_4	Dave
π_{E_2}	1	E_1	V_1	R'_{22}	E'_{22}	V'_{22}		$\vdots$	$\vdots$	$\vdots$
π_{E_3}	2	E_2	V_2	R'_{23}	E'_{23}	V'_{23}	$\pi_b \quad \pi_{gt}$	E''_K	pk_K	Nataly
π_{E_4}	3	E_3	V_3	$\vdots$	$\vdots$	$\vdots$	$\vdots$			
$\vdots$	$\vdots$	$\vdots$	$\vdots$	R'_{N1}	E'_{N1}	V'_{N1}	$\pi_b \quad \pi_{gt}$			
π_{E_M}	M	E_M	V_M	R'_{N2}	E'_{N2}	V'_{N2}	$\pi_b \quad \pi_{gt}$			
				R'_{N3}	E'_{N3}	V'_{N3}				

$$\pi_{b_{ij}} = \text{PoK} \left\{ b : E'_{ij} = \text{Com}((E'_{i1})^b) \wedge b \in \{0, 1\} \right\}$$

$$\pi_{gt_{ij}} = \text{PoK} \left\{ r : R_{i1}/R_{ij} = \text{Com}(g^r) \wedge r \in \{1, \dots, M\} \right\}$$

1 System Overview



4 Security Properties Across Roles

Property	Smartcard	Voter	Registrar	Collector	Tallier	BBoard
Eligibility Verifiability	●	○	●	○	○	○
Universal Verifiability	○	○	○	○	○	○
Individual Verifiability	●	●	○	○	○	○
Vote Anonymity	○	●	○	●	○	○
Coercion Resistance	●	○	○	●	○	○
Participation Privacy	●	○	○	●	○	○

Notation: ● – honest; ○ – may be corrupt; T(●) – threshold trust; two rows: alternative assumptions.

3 Lightweight Smartcard

Holds long-term key x ; only $\text{pk} = g^x$ is on the registration roll

Receives encrypted vote V , fresh blinding β , and nonce

Returns $\sigma = (E, \pi_E, \text{nonce})$

$$\pi_E = \text{PoK}^{(\text{nonce}, V)} \left\{ (\beta, x) : E = h^\beta g^x \right\}$$

The card only commits to x and proves knowledge of the opening — simple enough to certify.

Motivation and Contributions

Motivation

- Deniable vote updating gives coercion resistance, but needs a *cleansing* phase that discards superseded ballots.
- Making cleansing publicly verifiable, private and efficient is notoriously difficult.

Key Contributions

- Eligibility verifiability:** ballot shuffle matched to a blinded roll shuffle, ensuring at most one vote per voter;
- Universal verifiability:** group by blinded credential and prove each selected entry is the latest ballot;
- Everlasting privacy** of anonymity, participation, deniability: credentials appear only inside perfectly hiding commitments;
- Lightweight smartcard:** work reduced to a single Σ -proof.

5 Key Takeaways

- Verifiable by anyone**
Only the registration roll and smartcards are trusted.
- Everlasting privacy**
No future adversary, even a quantum one, can link votes to voters.
- Lightweight smartcard**
Signature-level work; the card never sees the plaintext vote.